MSc. Project Report: Survey on Equations over Finite Fields

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Abstract

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Preface

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Acknowledgements

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	Meaning
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	Meaning
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^\infty e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\ldots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\ldots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\ldots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\ldots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\ldots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\ldots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\ldots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\leq d}[X_1,\ldots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\ldots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \leq d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^\top C_f X$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \neq 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\ldots,d_n)$	$\#\left\{(t_1,\ldots,t_n) \in \prod_i [d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\ldots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\ldots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	**Meaning**
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))\, \chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha \in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))\, \chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h \in \Phi} \xi(h)\, \mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h \in \Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k \cdot \deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$

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Chapter 1

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Chapter A

Appendix

§ A.1 Continued Fractions over Finite Fields

Let $r_0, r_1 \in \mathbb{F}_q[X]$ with $r_1 \neq 0$. Applying the Euclidean algorithm yields a sequence

$$r_i = A_i r_{i+1} + r_{i+2}, \quad i = 0, 1, \ldots, s,$$

where $0 \leq \deg(r_{i+1}) < \deg(r_i)$ for $i = 1, \ldots, s$ and $r_{s+2} = 0$. Here $A_0, A_1, \ldots, A_s \in \mathbb{F}_q[X]$, with $A_1, \ldots, A_s$ of positive degree. This gives the *continued fraction expansion*

$$\frac{r_0}{r_1} = A_0 + \cfrac{1}{A_1 + \cfrac{1}{A_2 + \cdots}} =: [A_0, A_1, A_2, \ldots, A_s].$$

Define polynomials $P_i, Q_i \in \mathbb{F}_q[X]$ for $i = -1, 0, 1, \ldots, s$ by the recurrences

$$\begin{aligned} P_{-1} &= 1, \quad P_0 = A_0, \quad P_i = A_i P_{i-1} + P_{i-2} \quad (i = 1, \ldots, s), \\ Q_{-1} &= 0, \quad Q_0 = 1, \quad Q_i = A_i Q_{i-1} + Q_{i-2} \quad (i = 1, \ldots, s). \end{aligned}$$

It is clear that $\deg(P_{i-1}) < \deg(P_i)$ and $\deg(Q_{i-1}) < \deg(Q_i)$ for $i = 1, \ldots, s$. We extend the definition of degree by $\deg(f/g) = \deg(f) - \deg(g)$ for a rational function f/g.

Lemma A.1.1 Rational Approximants via Convergents

For any rational function ρ of nonnegative degree,

$$[A_0, A_1, \ldots, A_{i-1}, \rho] = \frac{\rho P_{i-1} + P_{i-2}}{\rho Q_{i-1} + Q_{i-2}} \quad \textit{for } i = 1, \ldots, s+1.$$

Proof: We proceed by induction on i. For $i = 1$ both sides equal $A_0 + \rho^{-1} = \rho^{-1}(A_0\rho + 1) = (\rho P_0 + P_{-1})/(\rho Q_0 + Q_{-1})$. Suppose the identity holds for some i, $1 \leq i \leq s$. Since $A_i + \rho^{-1}$ has positive degree, applying the inductive hypothesis gives

$$[A_0, \ldots, A_i, \rho] = \frac{(A_i + \rho^{-1})P_{i-1} + P_{i-2}}{(A_i + \rho^{-1})Q_{i-1} + Q_{i-2}} = \frac{\rho P_i + P_{i-1}}{\rho Q_i + Q_{i-1}},$$

using the recurrence relations. ■ ■

Corollary A.1.2 Convergents Formula

For $i = 0, 1, \ldots, s$ we have $[A_0, A_1, \ldots, A_i] = P_i/Q_i$.

Lemma A.1.3 Representation of r_0/r_1 via Convergents

For $i = 0, 1, \ldots, s$ we have

$$\frac{r_0}{r_1} = \frac{P_i + \beta_i P_{i-1}}{Q_i + \beta_i Q_{i-1}},$$

where $\beta_i = r_{i+2}/r_{i+1}$ is a rational function of negative degree.

Proof: We proceed by induction. For $i = 0$, $(P_0 + \beta_0 P_{-1})/(Q_0 + \beta_0 Q_{-1}) = (A_0 + r_2/r_1)/(1) = (A_0 r_1 + r_2)/r_1 = r_0/r_1$. The inductive step follows from the recurrences and $\beta_i^{-1} = A_{i+1} + \beta_{i+1}$. ■ ■

Lemma A.1.4 Determinant Identity for Convergents

For $i = 0, 1, \ldots, s$ we have $P_i Q_{i-1} - P_{i-1} Q_i = (-1)^{i-1}$.

Proof: Induction. For $i = 0$: $P_0 Q_{-1} - P_{-1} Q_0 = 0 \cdot 0 - 1 \cdot 1 = -1 = (-1)^{-1}$. Assuming the identity at i,

$$P_{i+1} Q_i - P_i Q_{i+1} = (A_{i+1} P_i + P_{i-1}) Q_i - P_i (A_{i+1} Q_i + Q_{i-1}) = -(P_i Q_{i-1} - P_{i-1} Q_i) = (-1)^i. \blacksquare$$

■

Corollary A.1.5 Coprimality of Convergents

For $i = 0, 1, \ldots, s$ we have $\gcd(P_i, Q_i) = 1$.

Proof: If $d_i = \gcd(P_i, Q_i)$ then Lemma A.1.4 shows d_i divides $(-1)^{i-1}$, so $d_i = 1$. ■ ■

§ A.2 A Product Identity for Roots of Unity

The following identity, used in the factorisation of lifted L-functions, expresses a product of linear factors twisted by roots of unity as a perfect power.

Lemma A.2.1 Roots-of-Unity Product Identity

Let $n, m \in \mathbb{N}$ and set $d = \gcd(m, n)$. Then

$$\prod_{t=1}^{n} \left(1 - e^{2\pi i m t/n} X\right) = \left(1 - X^{n/d}\right)^d.$$

Proof: Let $\omega = e^{2\pi i/n}$ be a primitive n-th root of unity, so the factors are $1 - \omega^{mt} X$ for $t = 1, \ldots, n$. Since $t = n$ gives $\omega^{mn} = 1 = \omega^0$, the product is the same as $\prod_{t=0}^{n-1} (1 - \omega^{mt} X)$.

Because $d = \gcd(m, n)$, the element ω^m is a primitive (n/d)-th root of unity. As t ranges over $\{0, 1, \ldots, n-1\}$, the values ω^{mt} cycle through the (n/d)-th roots of unity exactly d times each. Hence

$$\prod_{t=0}^{n-1} \left(1 - \omega^{mt} X\right) = \left(\prod_{k=0}^{n/d-1} \left(1 - (\omega^m)^k X\right)\right)^d.$$

Since ω^m is a primitive (n/d)-th root of unity, the inner product equals $1 - X^{n/d}$ by the standard factorisation of $1 - Y^N = \prod_{k=0}^{N-1}(1 - \zeta^k Y)$. Therefore the whole product equals $(1 - X^{n/d})^d$. ■ ■

§ A.3 Elementary Symmetric Polynomials

Waring's Formula expresses each power-sum symmetric polynomial $s_k = x_1^k + x_2^k + \cdots + x_n^k$ as an explicit polynomial in the elementary symmetric polynomials $\sigma_1, \ldots, \sigma_n$. It is the source of the closed-form reduction of lifted Kloosterman sums $K(\chi^{(s)}; a, b)$ to the ground-field sum $K(\chi; a, b)$ in Lemma **??**: applying Waring's formula to $\omega_1^s + \omega_2^s$ with $\sigma_1 = \omega_1 + \omega_2 = -K(\chi; a, b)$ and $\sigma_2 = \omega_1\omega_2 = q$ produces the binomial-coefficient expansion. We state the general identity here for reference.

Theorem A.3.1 Waring's Formula

Let $x_1, \ldots, x_n$ be indeterminates, let

$$\sigma_j = \sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} x_{i_1} x_{i_2} \cdots x_{i_j} \qquad (j = 1, \ldots, n)$$

be the elementary symmetric polynomials in $x_1, \ldots, x_n$, and let

$$s_k = x_1^k + x_2^k + \cdots + x_n^k$$

be the kth power-sum. Then for every integer $k \ge 1$,

$$s_k = \sum (-1)^{i_2 + i_4 + i_6 + \cdots} \cdot \frac{(i_1 + i_2 + \cdots + i_n - 1)! \cdot k}{i_1! \cdot i_2! \cdots i_n!} \cdot \sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n},$$

where the summation runs over all n-tuples $(i_1, \ldots, i_n)$ of non-negative integers satisfying $i_1 + 2i_2 + \cdots + ni_n = k$. Each coefficient of $\sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n}$ is an integer.

Theorem A.3.2

Suppose $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ is a symmetric polynomial. Let $\overline{X} = (X_1, \ldots, X_n)$. Then there exists a polynomial $g \in \mathbb{F}_q[Y_1, \ldots, Y_n]$ such that

$$f(X_1, \ldots, X_n) = g(\mathrm{ESym}_1(\overline{X}), \ldots, \mathrm{ESym}_n(\overline{X})).$$

Moreover,

(i) if $\deg_{X_i}(f) = d$ for all $i \in [n]$ then $\deg G = d$.

(ii) if $\deg f = D$ then for every monomial $Y_1^{d_1} \cdots Y_n^{d_n}$ of g with non-zero coefficient then $d_1 + 2 \cdot d_2 + \cdots + n \cdot d_n = D$.

§ A.4 Hasse Derivative

Definition A.4.1: Hasse Derivative

Let $\mathbb{F}$ be any field. Let $P \in \mathbb{F}[X_1, \ldots, X_n]$. Then $\bar{i} \in \mathbb{Z}_0^n$ Hasse derivative of P is denoted by $P^{(\bar{i})}(X_1, \ldots, X_n)$ or $H^{(\bar{i})}(P)$ which is the coefficient of $Z^{i_1} \cdots Z_n^{i_n} =: \overline{Z}^{\bar{i}}$ in the polynomial $P(X_1 + Z_1, \ldots, X_n + Z_n)$. Thus

$$P(X_1 + Z_1, \ldots, X_n + Z_n) = \sum_{\bar{i}} P^{(\bar{i})}(X_1, \ldots, X_n) \cdot \overline{Z}^{\bar{i}}$$

For univariate polynomials we immediately get a closed form formula of kth hasse derivative. Let $f \in \mathbb{F}[X]$ where $f(X) = \sum_{i=0}^{d} a_i X^i$. Then

$$f^{(k)}(X) = \sum_{i=0}^{d} \binom{k}{i} \cdot a_i X^{i-k}$$

From the definition we have the following observation

Observation A.1. *For any $f, f_1, f_2 \in \mathbb{F}_q[X_1, \ldots, X_n]$ and $\alpha \in \mathbb{F}$ and for any $\bar{k} \in \mathbb{Z}^n$ we have*

(i) $(f_1 + f_2)^{(\bar{k})}(X_1, \ldots, X_n) = f_1^{(\bar{k})}(X_1, \ldots, X_n) + f_2^{(\bar{k})}(X_1, \ldots, X_n)$

(ii) $(\alpha \cdot f)^{(\bar{k})}(X_1, \ldots, X_n) = \alpha \cdot f^{(\bar{k})}(X_1, \ldots, X_n)$.

Since throughout this report we only use hasse derivatives in the context of univariate polynomials we will only focus on univariate polynomials and see some properties of hasse derivatives. Here we take $\mathbb{F}$ to be any field.

Lemma A.4.1

(i) For $f_1, \ldots, f_t \in \mathbb{F}[X]$ then we have

$$(f_1 \cdot f_t)^{(n)}(X) = \sum_{\substack{n_1, \ldots, n_t \geq 0 \\ n_1 + \cdots n_t = n}} \prod_{i=1}^{t} f_i^{(n_i)}(X)$$

(ii) For any $\alpha \in \mathbb{F}$ we have

$$\left((X - c)^t\right)^{(n)} = \binom{t}{n} \cdot (X - c)^{t-n}.$$

(iii) For $0 \leq n \leq$ and $f, g \in \mathbb{F}[X]$ we have

$$\left(gf^t\right)^{(n)}(X) = \tilde{g} \cdot f^{t-n}(X)$$

where $\tilde{g} \in \mathbb{F}[X]$ with $\deg(\tilde{g}) \leq \deg(g) + n(\deg(f) - 1)$.

Proof:

(i) Since the hasse derivative operator is linear it is enough to show this property for monomials. So suppose $f_j(X) = X^{k_j}$ for all $j \in [t]$. Then $f_1 \cdots f_t = X^{\sum_{j=1}^{t} k_j}$. Therefore,

$$(f_1 \cdots f_t)^{(n)}(X) = \binom{\sum_{j=1}^{t} k_j}{t}, \qquad \prod_{j=1}^{t} f_j^{n_j}(X) = \prod_{j=1}^{t} \binom{k_j}{n_j} \text{ for all } n_j \in \mathbb{Z}_0, j \in [t]$$

So it is enough to show that

$$\binom{\sum_{j=1}^{t} k_j}{t} = \sum_{\substack{n_1, \ldots, n_t \geq 0 \\ n_1 + \cdots n_t = n}} \prod_{j=1}^{t} \binom{k_j}{n_j}$$

But this is true by comparing the coefficients of X^n in $(X+1)^{\sum_{j=1}^t k_j} = \prod_{j=1}^t (X+1)^{k_j}$. So we have the result.

(ii) Now take $f_j(X) = X - c$ for all $j \in [t]$. Then $f_j^{(n)}(X) = 1$ if $n = 1$ and 0 if $n > 1$. Therefore, if we use the first part on $f_1, \ldots, f_j$ in the sum on right-hand side the only terms which survives are when each n_j are either 0 or 1. The number of such terms are $\binom{t}{n}$ and each term is $(X-c)^{t-n}$.

(iii) Again by using the first part we get

$$(gf^t)^{(n)}(X) = \sum_{\substack{n_0, n_1, \ldots, n_t \geq 0 \\ n_0 + n_1 + \cdots + n_t = n}} g^{(n_0)}(X) \prod_{j=1}^{t} f^{(n_j)}(X)$$

■

Lemma A.4.2

Suppose $U(X), V(X) \in \mathbb{F}_q[X]$, $\deg(V) < q$, $i \geq 0$, *and*

$$U(X) = V(X) \cdot \Lambda^i(X).$$

Then for $0 \leq \ell < q$,

$$U^{(\ell)}(X) \equiv (-1)^i \cdot V^{(\ell - i)}(X) \pmod{\Lambda(X)}.$$

Theorem A.4.3

Suppose $\mathrm{char}(\mathbb{F}) = p$ *where* p *is a prime. Let* $Q(X, Y) \in \mathbb{F}[X, Y]$ *be a polynomial and define* $h(X) = Q\left(X, X^{p^s}\right)$ *for some* $s \in \mathbb{N}$. *Then for all* $0 \leq n < p^s$

$$h^{(n)}(X) = Q^{(n,0)}\left(X, X^{p^s}\right)$$

where $Q^{(i,0)}$ *is the* i^{th} *partial derivative of* Q *with respect to* X *where we consider* Q *as a polynomial of* $\mathbb{F}[Y][X]$.

Proof:

■

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